

Q:How to give read and write permissions using Unix commands. A very simple example is to create a text file with a C program, then use chmod.

```
Code: #include int main()
{ FILE *fp; fp = fopen("sample.txt", "w");
if (fp == NULL) { printf("Cannot open file\n");
return 1;
} fprintf(fp, "Hello, this is a test file.\n");
fclose(fp);
printf("File created successfully.\n");
return 0; }
```

Output:

```
sampa@oggy:~$ nano permission.c
sampa@oggy:~$ gcc permission.c -o permission
sampa@oggy:~$ ./permission
File created successfully.
sampa@oggy:~$ ls -l sample.txt
-rw-r--r-- 1 sampa sampa 28 Aug 17 04:48 sample.txt
```

Task of the Day

- 1) Give Read and Write only to the owner

Output:

```
sampa@oggy:~$ chmod 600 sample.txt
sampa@oggy:~$ ls -l sample.txt
-rw----- 1 sampa sampa 28 Aug 17 05:10 sample.txt
sampa@oggy:~$ |
```

- 2) Give Read and Write to the owner and group

Output:

```
sampa@oggy:~$ chmod 660 sample.txt
sampa@oggy:~$ ls -l sample.txt
-rw-rw---- 1 sampa sampa 28 Aug 17 05:10 sample.txt
sampa@oggy:~$ |
```

- 3) Only owner can read and write

Output:

```
sampa@oggy:~$ chmod 600 sample.txt
sampa@oggy:~$ ls -l sample.txt
-rw----- 1 sampa sampa 28 Aug 17 05:10 sample.txt
sampa@oggy:~$ |
```

4) Owner read/write, everyone else read-only

Output:

```
sampa@oggy:~$ chmod 644 sample.txt
sampa@oggy:~$ ls -l sample.txt
-rw-r--r-- 1 sampa sampa 28 Aug 17 05:10 sample.txt
sampa@oggy:~$ |
```

5) Owner can do everything, others can only read and execute

Output:

```
sampa@oggy:~$ chmod 766 sample.txt
sampa@oggy:~$ ls -l sample.txt
-rwxrw-rw- 1 sampa sampa 28 Aug 17 05:10 sample.txt
sampa@oggy:~$ |
```

6) Owner and group can read/write

Output:

```
sampa@oggy:~$ chmod 660 sample.txt
sampa@oggy:~$ ls -l sample.txt
-rw-rw---- 1 sampa sampa 28 Aug 17 05:10 sample.txt
sampa@oggy:~$ |
```

7) Only owner can execute

Output:

```
sampa@oggy:~$ chmod 100 sample.txt
sampa@oggy:~$ ls -l sample.txt
---x----- 1 sampa sampa 28 Aug 17 05:10 sample.txt
sampa@oggy:~$ |
```

Fill in the permission numbers for the following:

8) Owner: read + write; Group: nothing; Others: nothing → 600

9) Owner: read + write; Group: read; Others: read → 644

10) Owner: read + write + execute; Group: read + execute; Others: read + execute  
→ 755

11) Owner: read + write; Group: read + write; Others: nothing → 660

12) Everyone: read only → 444